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Dietary supplementation of coffee pulp extract enhances growth performance and intestinal morphology in broiler chicken

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ABSTRACT

Coffee is a renowned beverage derived from plants globally. In the coffee production process, coffee pulp is a by-product that is abundant in phenolic compounds. Therefore, this study aimed to assess the effect of coffee pulp extract (**CPE**) on growth performance, blood biochemistry, intestinal morphology, carcass characteristics, and meat quality of broiler chickens. A total of 160 one-day-old male Ross 308 broilers were randomly allocated to four treatments with five replicates and eight chicks per replicate in a completely randomized design. These four dietary treatments included a basal diet with CPE of 0 (**Control**), 250 (**CPE250**), 500 (**CPE500**), and 1,000 (**CPE1000**) mg/kg diet for 35 days. The results showed that the body weight (BW) of the CPE500 group was significantly higher ($P < 0.01$) compared to the other groups at 35 days. Moreover, CPE500 increased the average daily gain (**ADG**) ($P = 0.004$) and reduced the feed conversion ratio (**FCR**) ($P = 0.008$). No significant differences ($P > 0.05$) were observed in the blood biochemistry profile. In addition, the investigation on intestinal morphology showed that CPE supplementation enhanced villus height (**VH**) ($P = 0.004$), crypt depth (**CD**) ($P < 0.05$), and ratio of VH:CD ($P < 0.05$) in the duodenum. Dietary supplementation with CPE significantly increased the percentage of neck weight ($P < 0.05$) compared to the control groups. However, no significant effects of CPE supplementation were observed on the meat quality parameters of breast and thigh muscles, including pH, color, water-holding capacity, and tenderness ($P > 0.05$). A significant increase ($P < 0.05$) in thigh fat content was observed with CPE supplementation. In conclusion, CPE500 can improve the growth performance and intestinal morphology of broiler chickens despite the presence of antioxidants and anti-inflammatory agents. This suggests that coffee pulp biomass could potentially be used as an alternative feed additive from agricultural biomass in broiler production.

Introduction

The demand for affordable and high-quality broiler chicken meat is significantly increasing, particularly in developing countries with growing populations. This trend is expected to drive a substantial increase in per capita poultry meat consumption over the next decades (Kleyn and Ciacciariello, 2021; Kpomasse et al., 2021; Neima et al., 2023). Thailand is the key player in broiler production, which serves as a major supplier of chicken meat for domestic and global consumptions. In 2019, the country boasted 32,631 broiler chicken farms collectively housing 1,684.26 million birds and producing an impressive annual yield of 2.49 million tons of chicken meat (Kamporn et al., 2022).

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However, within the production line, diseases like colibacillosis, necrotic enteritis, and gangrenous dermatitis pose significant challenges, leading to substantial financial losses amounting to billions of dollars (Fancher et al., 2020; M'Sadeq et al., 2015). Feed additives are under scrutiny for their potential to boost broiler production, with a specific focus on enhancing the avian immune system for natural infection control (Phillips et al., 2023). These low-inclusion products, including antibiotics, phytogenics, acidifiers, prebiotics, and probiotics, are incorporated into diet formulations to improve the nutritional quality of feed and enhance animal performance and health. Antibiotics is a commonly uses in broiler chicken production to mitigate the risk of bacterial infections. This intervention can demonstrably improve growth performance and survival rates (Fancher et al., 2020; Huanhong et al., 2023). It is noteworthy that antibiotics can exhibit growth-promoting effects that are not solely attributable to their antimicrobial properties (Hassan et al., 2018). The underlying mechanism for this is likely linked to alterations in the gut microbiota, potentially enhancing nutrient absorption efficiency in chickens. However, the widespread use of antibiotics in livestock production has been implicated in the emergence and dissemination of antibiotic-resistant bacteria, a significant concern (Adedokun and Olojede, 2019; Vidovic and Vidovic, 2020). This antibiotic resistance poses a serious threat to poultry health and performance. It can have direct detrimental effects on birds and can also indirectly compromise their health by weakening their immune systems, making them more susceptible to secondary infections (Hayajneh et al., 2024). Furthermore, the misuse of antibiotics in poultry production not only jeopardizes poultry health but also poses a significant risk to human health. Resistant bacteria can transfer horizontally from animals to humans through the consumption of contaminated meat products, contributing to the emergence of antibiotic-resistant infections in people (Ahmed et al., 2023). This critical issue necessitates the implementation of effective monitoring and management strategies for antibiotic use in poultry production to minimize the development and dissemination of resistance. These concerns highlight the urgent need for effective monitoring and management of antibiotic use in poultry production to mitigate the development and spread of resistance (Marshall and Levy, 2011; Ullah et al., 2023).

Plant-derived bioactive compounds could exhibit intrinsic antimicrobial properties. These compounds contribute to the reduction of enteric pathogenic bacterial loads, potentially promoting gut health in chickens (Hayajneh, 2019). In addition, some plant compounds might interfere with the formation or structure of biofilms produced by bacteria to prevent themselves for the any interventional treatment (Mosaddad et al., 2023). Coffee (Rubiaceae), a globally consumed beverage has long been credited for its positive effects primarily linked to caffeine (Esquivel and Jiménez, 2012). Beyond its popularity, coffee holds immense economic importance as one of the most crucial crops. The two predominant species, *Coffea arabica* (arabica) and *Coffea canephora* (commonly known as robusta coffee), dominate the global market, with arabica representing 70 % and robusta 30 %. This underscores the vast economic relevance of coffee cultivation on a worldwide scale (Bastian et al., 2021). Coffee production generates significant by-products, such as husk, pulp, and silverskin, comprising almost 50 % of the fruit's dry weight (Hu et al., 2023). Despite being considered waste, coffee pulp contains 4-12 % protein, 1-2 % lipids, 6-10 % minerals, and 45-89 % total carbohydrates (Maxiselly et al., 2023). The pulp is also rich in valuably phytochemical compounds like caffeine, tannins, and phenolic acids like chlorogenic acids and caffeic acid, as well as alkaloids such as caffeine, along with essential nutrients (Franca and Oliveira, 2016). Although potentially toxic due to caffeine and tannins, its organic nature makes it an ideal substrate for antimicrobial processes, presenting opportunities for the production of value-added products (Pandey et al., 2000; Sangta et al., 2021). Additionally the present of phenolic compounds, known for their diverse physiological properties, including antioxidant, anti-inflammatory, anti-carcinogenic, anti-microbial, and anti-aging effects that play a crucial role in human health and are increasingly recognized as potent functional ingredients in foods, beverages, and cosmetics (Saewan et al., 2020). In biological nature, coffee pulp extracts (CPE) have been also reported to exhibit these essential biological activities (Kaisangsri et al., 2020). Given that CPE is primarily composed of phenolic compounds, which have demonstrated beneficial effects in layer chicken model (Qosimah et al., 2020), it is intriguing to investigate its potential impact on broiler production. This research aims to address a significant knowledge gap by investigating the potential of CPE as a novel feed additive to enhance broiler chicken performance. To date, limited research has explored the utilization of CPE in this context. This study will specifically examine the effects of CPE supplementation on key performance indicators, including growth performance, blood biochemistry, intestinal morphology, and meat quality. The findings of this research are expected to contribute significantly to our understanding of CPE's potential as a sustainable and effective feed supplement in poultry farming, potentially leading to improved animal health and productivity.

Material and methods

Coffee pulp extraction

The coffee pulp was obtained from Nan province, Thailand (19°30′04.8″N 100°39′53.3″E). The coffee pulp was sun-dried and ground into a fine powder before extraction. The powder was extracted using a 50 % ethanol solution via maceration at a 1:5 ratio (w/v) for 3 times, as described by Chalalai et al. (2015); Vijayalaxmi et al. (2015). This choice of ethanol concentration and extraction conditions was guided by its efficiency in dissolving bioactive compounds while maintaining their stability (Ramesh et al., 2024). Following extraction, the solvent was evaporated using a rotary evaporator (RC 600, KNF Neuberger AG, Balterswil, Switzerland) at 40 °C to concentrate the extract. The concentrated extract was then freeze-dried and stored at 4 °C to preserve its bioactivity until analysis and supplementation in chicken feed. The weight of the extract was recorded post-drying to ensure accurate dosing during feed formulation.

Chemical Characterization

Total phenolic content (TPC) was determined following the method of Iriondo-DeHond et al. (2019) with some modification. Briefly, the extract was mixed with 10 % of Folin-Ciocalteu reagent, and 6 % $NaCO_3$ solution. The mixture was incubated for 60 min at room temperature in the darkness. Then measured the absorbance at 765 nm using a microplate reader (SPECTRO star, BMG LABTECH, Offenburg, Germany). The total phenolic content was expressed as mg of gallic acid equivalents per g of extract (mg GAE/g extract).

The analysis of Total Flavonoid Content (TFC) was performed in accordance with Chen et al. (2021) method, with slight modifications implemented. Firstly, the ethanolic crude extract was mixed with distilled water and 5 % $NaNO_2$ incubated for 5 min. Then, 10 % $AlCl_3 \cdot 6H_2O$ solution was added, thereafter, incubated for 6 min and 1 M NaOH solution before being added with distilled water. Then the mixture was measured the absorbance at 510 nm using the microplate reader. The total flavonoid content was expressed as mg of catechin equivalents per g of extract (mg CE/g extract).

The methodology for phenolic compound analysis followed the approach outlined in Sangta et al. (2021). The concentration of phenolic compounds was determined using high performance liquid chromatography (HPLC) system (LC-20A, Shimadzu, Kyoto, Japan) equipped with a reverse phase C-18 column (250 × 4.6 mm, 5 m) (RESTEK, Bellefonte, PA) and an UV-vis diode array detector set to record at 280 nm for detecting the phenolic compounds. The following elution gradients were used to separate phenolic compounds using mobile phase: A included 5 % formic acid and distilled water, and B contained acetonitrile, formic acid and distilled water was in the ratio of 85:5:10. The total runtime was 20 min per sample: 0-4 min, 80 % A; 4-8 min, 25 % A; 8-10 min, 25 % A; 10-17 min, 80 % A; 17–20 min, 80 % A, at a flow rate of 1.0 mL/min. The standards and a sample of 10 μL were injected into the column. The retention time of the commercial standard phenolic compounds identified the peaks.

The 2,2-diphenyl-1-picrylhydracyl (DPPH) assay was determined using the method described by Segheto et al. (2018). A stock solution was prepared by 0.2 mM DPPH with methanol. For the assay, the extract was combined with DPPH and left to incubate in darkness at room temperature for 30 min. Following incubation, the absorbance was measured at a wavelength of 510 nm using the microplate reader. The results are presented as the half-maximal inhibitory concentration (IC_{50}) (mg/mL), indicating the minimum concentration of extract necessary to cause a 50 % reduction in DPPH absorbance.

The 2,2′-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) (ABTS) assay was determined using the method described by Nemzer et al. (2021). The stock solutions comprised a mixture of 7 mM ABST and potassium persulfate, left in a dark room for 12-16 h at room temperature. To create the working solution, the stock solution was combined with pH 7.4 phosphate buffered saline. During the assay, the extract was mixed with 0.2 mM ABTS working solution and incubated for 30 min. Subsequently, the absorbance was measured at a wavelength of 734 nm using the microplate reader. The results are presented as the IC_{50} (mg/mL), signifying the minimum concentration of the extract required to induce a 50 % reduction in ABTS absorbance.

The ferric reducing antioxidant power (FRAP) assay was determined following the methodology outlined by Liaqat et al. (2021). A fresh working FRAP solution was prepared by combining 300 mM acetate buffer, 10 mM 2,4,6-tris(2-pyridyl)-s-triazine (TPTZ) in 40 mM HCl, and 20 mM $FeCl_3$ in a ratio of 10:1:1. During the assay, the extract was mixed with the FRAP reagent, and the absorbance was measured at 593 nm using the microplate reader. The reported results represent the IC_{50} (mg/mL), indicating the minimum concentration of the extract necessary to produce a 50 % reduction in FRAP absorbance.

Ethical approval, animals and diets

The animal study protocol was conducted in strict accordance with the guidelines recommended and ethically approved by the Animal Ethics Committee, Faculty of Agriculture, Chiang Mai University with the protocol number AG02002/2567.

A total of 160 One-day-old male chicks (Ross 308) were purchased from CPF (Thailand) Public Company Limited, Thailand. The chickens were housed in 20 floor pen, each measuring 1 × 1.2 × 0.5 m, width, length, and high, separately and provided bedding material with clean rice husks. The chicks were accessed to clean water with nipple drinker. The light supplied 16 h per day and 8 h of darkness at an ambient temperature of not over 30°C. The basal diet was a commercial corn-soybean meal-based diet formulated to meet the nutritional requirements of broilers according to the National Research Council (1994). To achieve the intended amount of CPE addition, CPE was dissolved in 1 L of 80 % saline solution, sprayed onto the feed mixture while mixing with a feed mixer (Model MI1HP3V01, Siam Farm Service, Lampang, Thailand), and then dried at 60°C for 3 h. The nutritional composition of the diets was conducted following the guidelines outlined by the Association of Official Analytical Chemists (Cunniff and Washington, 1997). The chemical composition of the diets is presented in Table 1.

Table 1
Chemical composition of the commercial broiler diet.*

Proximate composition	Commercial diet	
	Starter diets (1-21 d)	Finisher diet (22-35 d)
Dry matter (% of feed)	90.23	90.38
Crude protein (% of DM)	20.34	19.66
Ether extract (% of DM)	4.79	3.98
Ash (% of DM)	6.55	6.89
Crude fiber (% of DM)	4.38	5.75
Gross energy (Cal/g)	3,928.73	3,911.60

* Commercial diets were purchased from a feed manufacturer. The chemical composition was verified through analytical testing, with results closely matching the values reported by the manufacturer.

Experimental design

The chicken was fed a basal diet and basal diet supplemented with CPE in the starter diet (1 − 21 d) and in the finisher diet (22–35 d). The chicks were randomly divided into four dietary treatments with five replicates and eight broilers per replicate with a completely randomized design (CRD). The four treatments were a basal diet (Control), a basal diet supplemented with CPE 250 mg/kg diet (CPE250), a basal diet supplemented with CPE 500 mg/kg diet (CPE500), and a basal diet supplemented with CPE 1000 mg/kg diet (CPE1000). Total experimental period was 35 d.

Data collection

The body weight of broiler in each group was recorded as 7, 14, 21, 28, and 35 d. The feed intake was recorded as every day in the experiment to calculate. The average daily feed intake (ADFI) and average daily gain (ADG) were calculated during the experiment period following the method of Alkhalf et al. (2010). The feed conversion ratio (FCR) was calculated as the feed intake per weight gain.

Blood biochemistry

On 35 d, 3 mL of blood was collected from the wing veins of randomly selected birds per treatment. The blood samples were quickly centrifuged at 3,000 rpm for 20 min using a Kokusan H-19α centrifuge (Kokusan, Saitama, Japan) to obtain the serum. The obtained serum was stored at −20°C until further analysis. For analysis, including total cholesterol, triglycerides, total protein, albumin, total bilirubin, direct bilirubin, aspartate-aminotransferase (AST), alanine-aminotransferase (ALT), and alkaline phosphate (ALP) (Longchuphon et al., 2024).

Intestinal morphology

At the end of experiment, three birds per group was randomly selected to collect small intestine. One inch of three sections of the small intestine were collected intestinal segments (duodenum, jejunum, and ileum). Then, normal saline was used to wash the samples thoroughly. The samples were preserved in 10 % neutral buffered formalin solution (Sharifi et al., 2012). After that, the samples were prepared for analysis through serial dehydration. The slides were prepared material with a rotary microtome. The sections were stained with Hematoxylin and Eosin (Daneshmand et al., 2017; Ji et al., 2019). Villus height (VH) and crypt depth (CD) were measured d at 40× magnification using a digital camera (Motic MC2000) connected to a compound microscope (CX21, Olympus Corporation, Tokyo, Japan). Villus height was measured from the tip of the villus to the villus-crypt junction, while cryptal depth was measured as the depth between two adjacent villi (Shini et al., 2020).

Carcass characteristics, meat quality and chemical analysis

By the end of the experiment (35 d), a total of ten birds per replicate were randomly selected. The chickens were fasted for 12 h (Chaiwang et al., 2023). After the fasting period, their body weights were recorded, and the birds were euthanized, followed by exsanguination. Subsequently, the birds were semi-scalded at 65°C for 1 min and defeathered. After the removal of feathers, viscera, abdominal fat, shanks, and neck, the weights of the eviscerated hot carcasses were measured. After defeathering, the chickens were weighed, and the weight were expressed as a percentage of the live killing weight and the feather-free weight. The wings, thighs, drumsticks, legs, breasts, skeletons, head and neck were cut using a scalpel. These were weighed and expressed as a percentage of the pre-slaughter weight. The weights of the internal organs, including the liver, spleen, gizzard, heart, and intestine, were recorded and expressed as a percentage of the killing weight and relative organ weight.

Breast and thigh inner portions have their color measured separately using the CIELAB scale for lightness (L*), redness (a*), and yellowness (b*) using a Minolta Chroma Meter CR-5 (Minolta, Osaka, Japan). The pH of the meat was measured using a Testo 205 T-bar handheld pH meter (Testo, Hampshire, UK) at 45 min and 24 h postmortem. After carcass evaluation, ten skinless breast meat samples per treatment were stored in a refrigerator before being tested for water-holding capacity, including drip loss, thaw loss, and cooking loss according to Chaiwang et al. (2023); Norkeaw et al. (2023). Meat samples were cut into pieces ($1 \times 1 \times 3$ cm). Additionally, the sample were cut into small pieces using an electric saw and immediately ground in a meat grinder (Retsch SM 200, Haan, Germany) using an 8 mm diameter mesh plate (Sosnówka-Czajka et al., 2023). Then, the breast and thigh meat were homogenized in an industrial mixer. Approximately 100 g of the sample were placed in sealed plastic bags and stored at −20°C for later chemical analysis. Proximate analysis of meat was conducted using procedures of AOAC (1997).

Statistical analysis

All experimental data were analyzed using analysis of variance (ANOVA) procedure of SPSS Statistic Software v.29.0 (SPSS Inc., Chicago, IL). The differences between the treatments were determined using Duncan's multiple range test. The data was shown as means with the standard error of the total mean (SEM) included. A probability level of $P < 0.05$ was considered as statistically significant.

Results

The extraction of CPE using ethanol as a solvent resulted in a yield of approximately 15 %. Subsequent analysis of the extracted substance revealed the TPC value around 56 mg GAE/g extract and TFC value at approximately 70 mg CE/g extract. Moreover, quantitatively analysis of the phenolic compounds in CPE revealed that caffeic acid and chlorogenic acid, the main constituents of coffee, both had a concentration of 0.98 mg/g extract, especially the catechin had the highest concentration, approximately 189 mg/g of extract. Furthermore, antioxidant activity was assessed through DPPH, ABTS, and FRAP assays, and the results, presented as IC_{50}, were found to be 0.25, 0.64, and 0.42 mg/mL, as summarized in Table 2.

The effects of CPE supplementation on broiler performance at different experimental periods is shown in Table 3. The was no significant difference ($P > 0.05$) on ADFI in each period among the treatments. BWG and ADG was significantly ($P < 0.05$) higher in the CPE500 group than in another group at 35 d. In addition, broiler fed with CPE500 had a lower FCR ($P = 0.008$).

Table 4 summarizes the effects of CPE on blood biochemistry at 35 d of age. According to the statistical analysis, the CPE supplement groups had no significant ($P > 0.05$) effects on the blood biochemical parameters including total cholesterol, triglycerides, total protein, albumin, globulin, total bilirubin, direct bilirubin, indirect bilirubin, AST, ALT, and ALP there were no significant differences in blood biochemistry of broiler chickens. This meant that the blood parameters were all within normal ranges.

Effects of crude extract of coffee pulp on intestinal morphology of broiler chickens are summarized in Table 5 and Fig. 1. Compared with the control group, the duodenum in the CPE500 statistically increased the VH ($P = 0.004$). Moreover, supplementation of CPE trended to increase the VH of jejunum and ileum compared with that of the control group. However, a significant difference ($P < 0.05$) was observed for crypt depth and VH:CD ratio as supplementing CPE increased crypt depth and decreased VH:CD ratio compared to that of the control group.

Table 2
The phytochemical of the coffee pulp extract.

Variables	coffee pulp extract
Yield, %	14.34 ± 0.56
Phytochemical (mg/g extract)	
Total phenolic content	56.42 ± 2.30
Total flavonoid content	70.73 ± 8.01
Polyphenol composition (mg/g extract)	
Catechin	189.19 ± 5.32
Epicatechin	38.38 ± 0.08
Gallic acid	3.65 ± 0.04
Naringin	1.62 ± 0.04
Caffeic acid	0.98 ± 0.08
Chlorogenic acid	0.98 ± 0.001
Quercetin	0.61 ± 0.16
ρ-Coumeric	0.50 ± 0.01
Rosmarinic	0.20 ± 0.02
o-Coumeric	0.01 ± 0.01
Antioxidant activities (IC_{50}), mg/mL	
ABTS	0.25 ± 0.05
DPPH	0.64 ± 0.09
FRAP	0.42 ± 0.01

Data are expressed as mean ± standard deviation.

The effects of CPE on carcass trait and internal organ of broiler chickens are presented in Table 6. CPE supplementation has no effect on live weight, percentage of carcass yield and internal organ percentage ($P > 0.05$). While in cut-up parts, the addition of CPE in diet increased the percentage of neck weight ($P < 0.05$) compared with the control groups.

The effects of CPE on meat quality of broiler breast are shown in Table 7. In the breast section, the pH value, meat color, shear force and chemical composition did not differ significantly between the treatment and control groups ($P > 0.05$).

As shown in Table 8, the effects of CPE on the meat quality of broiler thighs. Meat quality from the broilers fed with CPE supplements had no effect on pH value, meat color, water holding, shear force, and chemical composition. However, the CPE-supplemented group had a lower fat content in the thigh ($P < 0.05$) compared to the control group. Additionally, the CPE levels were lower in chickens fed with the supplemented diet compared to those fed the control diet.

Discussion

Intricate relationship between the choice of coffee variety, extraction solvent, and the subsequent levels of the phytochemical analysis has been elucidated by the variability in these parameters across different experimental conditions and methods. In the work of Geremu et al. (2016), the extraction of the pulps of the arabica coffee varieties; Ababuna, 741, Dessu, and 74110 using acetone, ethanol, and methanol solvents showed that the choice of solvent did not significantly affect the TPC and antioxidant as determined by the DPPH assay results. When ethanol was used, the TPCs in these verities were determined to be around 500-1,600 mg/g, respectively. In addition, the extraction yield and antioxidant activity of plant extracts are heavily influenced by the polarity of the solvents used. Solvent polarity plays a crucial role in determining both the qualitative and quantitative profiles of the extracted antioxidant compounds. Ethanol and methanol, especially when mixed with water, are commonly employed and are known to achieve the highest extraction yields (Ramesh et al., 2024; Vijayalaxmi et al., 2015). Simultaneously, the DPPH antioxidant quantities were found to be 2.4- 15.4 mg/mL, respectively. However, our experimental results indicated comparatively lower values in both cases. This suggests that, in fact, the choice of coffee variety has a discernible impact on the contents of bioactive ingredients and the biological properties.

Table 3
Growth performance of broiler chickens at different experimental week.

Variables	Control	CPE250	CPE500	CPE1000	SEM	*P*-value
BW; g						
Initial BW	43.74	43.85	43.31	44.05	0.13	0.257
7 d	146.11	148.84	146.66	147.68	0.67	0.551
14 d	390.42	392.42	389.63	397.50	1.20	0.055
21 d	733.05[b]	754.29[ab]	763.25[a]	763.67[a]	4.73	0.040
28 d	1,270.59	1,282.25	1,307.75	1,289.42	5.73	0.115
35 d	1,841.59[b]	1,847.05[b]	1,930.21[a]	1,869.21[b]	11.87	0.004
ADG; g/day						
1 – 7 d	14.68	14.84	14.81	14.86	0.08	0.907
8 – 14 d	34.79	34.70	34.96	35.63	0.16	0.130
15–21 d	51.50	52.27	53.34	53.73	0.40	0.187
22–28 d	75.53	74.29	77.53	76.82	0.67	0.373
29–35 d	79.82[b]	80.69[b]	88.57[a]	83.64[ab]	1.24	0.021
ADFI; g/hen/day						
1 – 7 d	22.01	22.54	22.42	22.60	0.12	0.311
8 – 14 d	56.07	56.78	56.18	56.34	0.17	0.751
15–21 d	100.43	101.01	100.95	100.67	0.25	0.864
22–28 d	140.33	141.31	146.40	142.65	0.92	0.066
29–35 d	167.37	163.42	172.63	172.67	1.63	0.460
FCR						
1 – 7 d	1.50	1.52	1.51	1.52	0.001	0.876
8 – 14 d	1.61	1.61	1.61	1.58	0.007	0.591
15–21 d	1.95	1.93	1.89	1.87	0.02	0.262
22–28 d	1.86	1.90	1.89	1.86	0.02	0.796
29–35 d	2.09[b]	2.06[b]	1.94[a]	2.06[b]	0.02	0.008

Abbreviations: Control, basal diet supplemented with CPE 0 mg/kg diet; CPE250, basal diet supplemented with CPE 250 mg/kg diet; CPE500, basal diet supplemented with CPE 500 mg/kg diet; CPE1000, basal diet supplemented with CPE 1,000 mg/kg diet. BW, body weight; ADG, average daily gain; ADFI, average daily feed intake; FCR, feed conversion ratio.

[a,b] Means within a row with different superscripts differ significantly ($P < 0.05$).

Table 4
Effects of coffee pulp extract (CPE) on blood biochemistry of broiler chickens.

Variables	Control	CPE250	CPE500	CPE1000	SEM	*P*-value
Total cholesterol (mg/dL)	92.33	108.00	107.67	123.00	5.23	0.243
Triglycerides (mg/dL)	48.33	47.33	54.67	47.00	2.67	0.772
Total protein (mg/dL)	3.13	3.13	2.97	3.13	0.08	0.869
Albumin (mg/dL)	1.07	1.20	1.07	1.20	0.03	0.281
Globulin (mg/dL)	2.07	1.93	1.90	1.93	0.07	0.858
Total bilirubin (mg/dL)	0.06	0.04	0.02	0.05	0.007	0.421
Direct bilirubin (mg/dL)	0.03	0.02	0.02	0.02	0.003	0.463
Indirect bilirubin (mg/dL)	0.03	0.03	0.01	0.03	0.006	0.584
AST (U/L)	298.67	302.67	293.00	260.33	11.33	0.600
ALT (U/L)	nd[1]	nd	nd	nd		
ALP (U/L)	1,361.30	1,391.00	1,544.70	1,780.30	72.07	0.135

Abbreviations: Control, basal diet supplemented with CPE 0 mg/kg diet; CPE250, basal diet supplemented with CPE 250 mg/kg diet; CPE500, basal diet supplemented with CPE 500 mg/kg diet; CPE1000, basal diet supplemented with CPE 1,000 mg/kg diet. AST, aspartate-aminotransferase; ALT, alanine-aminotransferase; ALP, alkaline phosphate.

[1] nd, not detected.

Table 5
Effects of coffee pulp extract (CPE) on intestinal morphology of broiler chickens.

Variables	Control	CPE250	CPE500	CPE1000	SEM	*P*-value
Duodenum						
VH (μm)	1,199.78[b]	1,214.81[b]	1,302.54[a]	1,225.41[b]	11.46	0.004
CD (μm)	170.98[b]	213.77[a]	189.67[ab]	188.10[ab]	5.24	0.031
VH:CD	7.06[a]	5.83[b]	6.98[a]	6.79[ab]	0.17	0.049
Jejunum						
VH (μm)	1,331.15	1,355.46	1,400.74	1,338.31	10.88	0.098
CD (μm)	115.26	116.42	113.51	128.05	3.58	0.483
VH:CD	11.99	11.99	12.65	10.83	0.36	0.348
Ileum						
VH (μm)	835.92	914.31	972.78	903.65	18.04	0.057
CD (μm)	105.02	113.57	103.52	101.10	3.00	0.500
VH:CD	8.14	8.28	9.54	9.15	0.26	0.153

Abbreviations: Control, basal diet supplemented with CPE 0 mg/kg diet; CPE250, basal diet supplemented with CPE 250 mg/kg diet; CPE500, basal diet supplemented with CPE 500 mg/kg diet; CPE1000, basal diet supplemented with CPE 1,000 mg/kg diet. VH, Villus height; CD, crypt depth; VH:CD, the ratio of villus height to crypt depth.

[a,b] Means within a row with different superscripts differ significantly ($P < 0.05$).

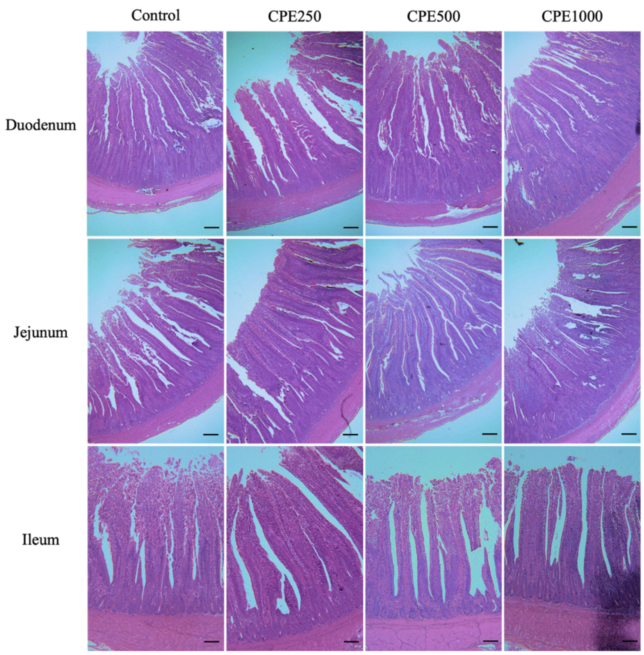

Fig. 1. Broiler intestines after diet supplementation of CPE based on hematoxylin and eosin staining observed under 40x magnification. Control, basal diet supplemented with CPE 0 mg/kg diet; CPE250, basal diet supplemented with CPE 250 mg/kg diet; CPE500, basal diet supplemented with CPE 500 mg/kg diet; CPE1000, basal diet supplemented with CPE 1,000 mg/kg diet.
Scale bar indicates 200 µm.

Table 6
Effects of coffee pulp extract (CPE) on carcass trait and internal organ of broiler chickens.

Variables	Control	CPE250	CPE500	CPE1000	SEM	*P*-value
Live weight, kg	1.84^{b}	1.85^{b}	1.93^{a}	1.87^{b}	0.11	0.004
Carcass yield, %	78.59	78.52	77.96	78.37	0.15	0.463
Carcass compositions (% of live weight)						
Head	2.71	2.56	2.74	2.55	0.06	0.508
Neck	4.03^{c}	4.08^{bc}	4.40^{ab}	4.48^{a}	0.06	0.018
Feet	4.01	4.12	3.92	4.12	0.04	0.265
Breast	21.27	22.17	21.12	19.75	0.63	0.522
Thigh	10.78	10.31	10.79	10.55	0.17	0.713
Drumstick	10.79	11.08	11.07	10.95	0.09	0.695
Wing	8.04	7.92	8.11	8.22	0.09	0.707
Skeleton	21.28	21.43	20.48	21.31	0.31	0.716
Internal organ (% of carcass weight)						
Liver	2.19	2.20	2.23	2.11	0.04	0.726
Gizzard	2.79	2.71	2.74	2.88	0.07	0.838
Heart	0.51	0.50	0.53	0.52	0.009	0.736
Spleen	0.11	0.11	0.12	0.10	0.006	0.622
Intestine	4.26	4.44	4.60	4.33	0.09	0.555

Abbreviations: Control, basal diet supplemented with CPE 0 mg/kg diet; CPE250, basal diet supplemented with CPE 250 mg/kg diet; CPE500, basal diet supplemented with CPE 500 mg/kg diet; CPE1000, basal diet supplemented with CPE 1,000 mg/kg diet.
[a,b,c] Means within a row with different superscripts differ significantly ($P < 0.05$).

Comparing these findings to research by Chalalai et al. (2015), which involved the extraction of the pulp from Arabica coffee variety in Thailand using ethanol, the TPC was reported as 0.13 mg GAE/g, and the antioxidants by the DPPH and FRAP assays were found at 2.22 mmol Trolox equivalent/100 g, and 9.17 mmoL Ascorbic acid equivalent/100 g. Additionally, Sangta et al. (2021) analyzed the polyphenolic contents and antioxidant potentials in methanol and dichloromethane in the Arabica coffee pulps. The methanol extract exhibited TPC of 1.03 µg/g dried sample and an impressive DPPH antioxidant activity result of 99.44 %. In contrast, the dichloromethane extract showed no detectable TPC and TFC, with a low percentage of antioxidant inhibition.

Table 7
Effects of coffee pulp extract (CPE) on meat quality of broiler breast at 35 d of ages.

Variables	Control	CPE250	CPE500	CPE1000	SEM	*P*-value
pH at 45 min	6.64	6.53	6.74	6.53	0.04	0.101
pH at 24 h	5.85	5.77	5.82	5.87	0.02	0.366
Meat color						
Lightness (L*)	61.89	57.86	60.46	59.52	0.81	0.360
Redness (a*)	−0.28	−0.23	−0.66	−0.48	0.11	0.203
Yellowness (b*)	11.33	11.49	11.16	10.65	0.27	0.081
Water holding						
Drip loss, %	3.36	3.04	2.88	2.65	0.14	0.321
Thawing loss, %	3.38	3.26	3.67	4.35	0.39	0.772
Cook loss, %	17.88	17.49	19.54	18.45	0.60	0.663
Shear force, N	30.29	32.23	31.44	31.82	0.53	0.611
Proximate composition, %						
Moisture	75.59	75.10	75.54	75.19	0.13	0.486
Crude protein[1]	22.35	22.41	22.07	22.71	0.11	0.269
Ether extract[1]	1.07	1.34	1.19	1.26	0.07	0.582
Ash[1]	1.47	1.46	1.49	1.44	0.02	0.642

Abbreviations: Control, basal diet supplemented with CPE 0 mg/kg diet; CPE250, basal diet supplemented with CPE 250 mg/kg diet; CPE500, basal diet supplemented with CPE 500 mg/kg diet; CPE1000, basal diet supplemented with CPE 1,000 mg/kg diet.
[1] Proximate analysis present as dry matter basis.

Table 8
Effects of coffee pulp extract (CPE) on meat quality of broiler thigh at 35 d of ages.

Variables	Control	CPE250	CPE500	CPE1000	SEM	*P*-value
pH at 45 min	6.59	6.60	6.56	6.53	0.04	0.952
pH at 24 h	6.53	6.47	6.47	6.50	0.03	0.834
Meat color						
Lightness (L*)	57.39	57.38	58.54	56.90	0.32	0.309
Redness (a*)	0.56	0.46	0.31	0.32	0.10	0.781
Yellowness (b*)	8.03	7.86	8.49	8.52	0.28	0.790
Water holding						
Drip loss, %	2.26	2.45	2.50	2.31	0.08	0.657
Thawing loss, %	2.69	3.21	2.99	2.68	0.26	0.865
Cook loss, %	12.93	18.52	12.26	17.01	1.14	0.143
Shear force, N	17.03	16.24	18.67	19.69	0.72	0.328
Proximate composition[1], %						
Moisture	74.87	74.71	75.04	75.53	0.14	0.202
Crude protein[1]	18.30	18.61	18.21	18.59	0.16	0.777
Ether extract[1]	5.42^{a}	5.31^{ab}	4.73^{b}	4.72^{b}	0.11	0.029
Ash[1]	1.15	1.12	1.16	1.16	0.02	0.767

Abbreviations: Control, basal diet supplemented with CPE 0 mg/kg diet; CPE250, basal diet supplemented with CPE 250 mg/kg diet; CPE500, basal diet supplemented with CPE 500 mg/kg diet; CPE1000, basal diet supplemented with CPE 1,000 mg/kg diet.
[1] Proximate analysis present as dry matter basis.
[a,b] Means within a row with different superscripts differ significantly ($P < 0.05$).

The comprehensive analysis that provided valuable insights into the diverse composition of the polyphenols in coffee pulp was conducted in many research studies (Muangsanguan et al., 2023; Ramirez-Martinez, 1988; Sangta et al., 2021). The methanol extraction of Arabica coffee peels highlighted the prevalence of caffeic acid, reaching a maximum of 68.05 mg/g extract, while gallocatechin gallate was detected at the lowest concentration, measuring 0.12 mg/g extract (Sangta et al., 2021). Muangsanguan et al. (2023) utilized supercritical fluid extraction for Arabica coffee pulp; the polyphenolic content exhibited variability, with caffeic acid ranging from 4.07 to 19.49 mg/g dry weight and catechin gallate levels between 0.02 and 0.26 mg/g dry weight. Expanding the scope to include the pulp of 12 coffee cultivars in experiment of

Ramirez-Martinez (1988), the primary polyphenolic components were identified as chlorogenic acid (5-caffeoylquinic acid), constituting 42.2 %, followed by epicatechin at 21.4 %, isochlorogenic acid I at 5.7 %, and isochlorogenic acid II at 19.3 %. Our result indicates the rich polyphenolic composition and notable antioxidant properties of the ethanol extract derived from Robusta coffee pulp, with catechin emerging as a major component alongside chlorogenic acid, epicatechin, and caffeic acid, highlighting the diverse bioactive profile. Catechin, a natural flavonoid, plays a vital role in promoting broiler chicken health by mitigating oxidative stress through its potent antioxidant properties, enhancing immune function, and supporting gut health through its antimicrobial effects, making it a promising natural additive for improving broiler performance and overall well-being but also contribute to the alternative utilization of coffee biomass (Jelveh et al., 2022; Mahlake et al., 2022).

Coffee by-product is reported to be a source of polyphenols that have been reported to have powerful antioxidants that help reduce oxidative stress in animals (Salinas-Rios et al., 2015) and no negative effect on growth (Ko et al., 2012). The utilization of natural phenolic compounds in poultry nutrition demonstrating crucial biological properties. These include promoting growth, providing antioxidant effects, exhibiting antibacterial properties, and contributing to immunomodulation (Awodoyin et al., 2024; Saeed et al., 2019). The results of the most recent study highlight the relevance of adding high-quality botanical extracts to broiler chicken diets to improve feed efficacy and nutritional content, as feed quality has an important effect on gut health and overall performance. It supports the potential of botanical compounds as natural alternatives to synthetic additives in poultry farming (Anwar et al., 2023; Rehman et al., 2023). Several previous studies have shown that coffee extracts containing phenolic compounds (chlorogenic compounds) as components in broiler nutrition can contribute to increased BW and weight gain, along with reduced FCR (Liu et al., 2023). In another study, Mendes et al. (2013) reported that commercial coffee bean powder had no significant effect on feed intake. A study by Ashour et al. (2020) demonstrated that supplementation with green coffee bean powder significantly improved FCR in broiler chickens. This improvement is likely attributed to the synergistic effects of phenolic compounds in the coffee, which enhanced feed intake and overall performance. Green coffee bean powder supplementation has been reported to promote weight gain and decreases FCR (Hosseini-Vashan et al., 2023). Similar to present results are consistent with previous findings in broilers, which showed broilers fed CPE linearly increased ADG during the full growth period, and there is a potential linear trend of increase in BW at CPE levels on 22−35 d.

Blood serves as a valuable medium for reliably assessing the physiological and health status of animals (Erinle et al., 2022). In this study, supplementing feed additives had no effect on liver function or blood protein. There was no significant difference in total cholesterol, triglycerides, total protein, albumin, globulin, bilirubin (total, direct, and indirect), AST, ALT, or ALP levels among dietary treatments. Similarly, Awodoyin et al. (2024) who reported that graded levels of black coffee bean powder had no impact on the blood protein profile. Serum blood parameters are commonly used to assess the nutritional, pathological, and physiological status of chicks, reflecting the effects of diets supplemented with feed additives.

The intestinal villi are the primary site of nutrient absorption, and the integrity of the intestinal morphology has a significant impact on animal health. Increased villus height enhances the absorptive area, promoting nutrient uptake and optimizing production performance in broiler chickens (Gao et al., 2024; Lee et al., 2023). This agree with Ashour et al. (2020) reported that phytochemicals in natural additives are absorbed in the intestine and quickly metabolized. This results in improved absorptive function of the small intestine. In contrast, Viveros et al. (2011) report reduced villus height and crypt depth with grape polyphenol supplementation. In this study, the CPE-supplemented group exhibited significantly increased VH in the duodenum. Additionally, this group showed a trend towards significantly higher VH in both the jejunum and ileum. Moreover, despite a reduced VH:CD ratio in CPE-fed chickens. This may be due to the antioxidant and anti-inflammatory properties of CPE polyphenols, which may inhibit gastrointestinal pathogens and increase the nutrient absorption surface. It has been reported that natural polyphenol plant extracts can modify intestinal inflammation (Romier et al., 2009). Previous studies in mice fed coffee pulp polyphenol extract alleviated brain inflammation in colitis mice by inhibiting the NF-κB signaling pathway and modulating gut microbiota (Li et al., 2022). Therefore, the potential of green bean coffee extract has potential anti-inflammatory and anti-oxidant (Rosyidi et al., 2020). Our results demonstrated that the supplementation of CPE improved intestinal health and weight gain in broiler chickens. These effects can be attributed to its bioactive compounds, particularly phenolic compounds, which possess antioxidant, antimicrobial, and prebiotic-like properties (Chalalai et al., 2015; Duangjai et al., 2016; Garcia-Alonso et al., 2023). These compounds help promote a balanced gut microbiota, enhance nutrient absorption by improving intestinal integrity, and reduce oxidative stress, collectively contributing to improved growth performance and feed efficiency.

Additionally, our result demonstrated that supplementation of CPE did not affect the percentage of carcass yield or internal organ. Similarly, the previous study reported that carcass yield and relative organ weight were not affected by the supplementation of natural antioxidants (Marzoni et al., 2014). even though the body weight of the chickens increased. In term of chicken meat quality, important markers of include pH, color, water-holding capacity, and tenderness (Yang et al., 2023). This study suggested that supplementation with CPE had no effect on meat quality parameters, including pH, color value, water-holding capacity, and tenderness. Moreover, meat color, which is the outward manifestation of a variety of physiological and biochemical changes in muscles, is the most intuitive index of meat quality (Pereira Farias et al., 2019). We observed a trend towards a decrease in yellowness (b*) in the meat color of the breast ($P = 0.081$). Similarly, Mazur-Kuśnirek et al. (2019) had reported that broilers fed with polyphenol-enriched diets had a lower contribution of yellowness (b*) in the breast muscles. Additionally, CPE supplementation altered meat nutrition, especially fat, with fat accumulation in the breast and thigh muscles, which may be due to the influence of diet on fat accumulation in meat. It was reported that the energy content of the ration and feed intake are key factors influencing lipid accumulation and contributing to increased fat deposition in muscle tissue (Starčević et al., 2015). As reported by Hudák et al. (2021), the increase in lipids due to lipid oxidation was also affected by the broiler diet modification.

Conclusion

Dietary supplementation with 500 mg/kg of coffee pulp extract (CPE) significantly improved growth performance and enhanced intestinal structure in broiler chickens. This was evidenced by an increase in villus height and the villus height-to-crypt depth ratio in the duodenum. Notably, no significant effects were observed on blood biochemistry, carcass yield, or meat quality in the chickens fed the CPE-supplemented diet. These findings suggest that dietary inclusion of 500 mg/kg CPE can effectively support intestinal health and promote growth performance in broiler chickens without any adverse effects on meat quality.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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